

《计算机网络》作业

第六次作业：（英文教材第一章 9, 第四章 6,13,16,17,38）

1-9 (4.2.1 ALOHA). A disadvantage of a broadcast subnet is the capacity wasted when multiple hosts attempt to access the channel at the same time. As a simplistic example, suppose that time is divided into discrete slots, with each of the n hosts attempting to use the channel with probability p during each slot. What fraction of the slots will be wasted due to collisions?

6. (4.2.2 Carrier Sense Multiple Access Protocols) What is the length of a contention slot in CSMA/CD for (a) a 2-km twin-lead cable (signal propagation speed is 82% of the signal propagation speed in vacuum)?, and (b) a 40-km multimode fiber optic cable (signal propagation speed is 65% of the signal propagation speed in vacuum)?

13. (4.3.1 Classical Ethernet) What is the baud rate of classic 10-Mbps Ethernet?

16. (4.3.2 Classical Ethernet MAC Sublayer Protocol) Consider building a CSMA/CD network running at 1 Gbps over a 1-km cable with no repeaters. The signal speed in the cable is 200,000 km/sec. What is the minimum frame size?

17. (4.3.2 Classical Ethernet MAC Sublayer Protocol) An IP packet to be transmitted by Ethernet is 60 bytes long, including all its headers. If LLC is not in use, is padding needed in the Ethernet frame, and if so, how many bytes?

38. (4.3.4 Switched Ethernet) Store-and-forward switches have an advantage over cut-through switches with respect to damaged frames. Explain what it is.